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UNIT 3 PHYSICAL CHEMISTRY

In a world where the population has more than trebled in the last 100 years, fighting against hunger is an extremely important challenge. At the beginning of the 20th century scientists developed industrial processes to make artificial fertilisers that can be used to increase crop yields. In order to develop these processes, chemists and chemical engineers from around the world have had to understand the principles of energy changes, rates of reaction and equilibrium. The world's resources are limited and it is vital that these processes are as efficient and environmentally friendly as possible. This is also the work of scientists.



▲ Figure 19.1 Burning fuel produces enough energy to launch a rocket.

19 ENERGETICS

Some chemical reactions produce heat. Others need to be heated constantly to make them occur at all. This chapter explores some examples of both kinds of reaction and examines how energy changes during reactions can be measured by experiments or calculated using bond energies.

LEARNING OBJECTIVES

- Know that chemical reactions in which heat energy is given out are described as exothermic, and those in which heat energy is taken in are described as endothermic
- Describe simple calorimetry experiments for reactions such as combustion, displacement, dissolving and neutralisation
- Calculate the heat energy change from a measured temperature change using the expression $Q = mc\Delta T$
- Calculate the molar enthalpy change (ΔH) from the heat energy change, Q
- Practical: Investigate the temperature changes accompanying some of the following types of change:
  - salts dissolving in water
  - neutralisation reactions
  - displacement reactions
  - combustion reactions

CHEMISTRY ONLY

- Draw and explain energy level diagrams to represent exothermic and endothermic reactions.
- Know that bond breaking is an endothermic process and that bond making is an exothermic process.
- Use bond energies to calculate the enthalpy change during a chemical reaction.

EXOTHERMIC REACTIONS

Some chemical reactions give out energy in the form of heat. A reaction that *gives out heat to the surroundings* is said to be **exothermic**. If you are holding a test-tube in which an exothermic reaction is occurring, you will notice that the test-tube *gets warmer*.

An example of an exothermic reaction is adding water to calcium oxide. If you add water to solid calcium oxide, the heat produced is enough to boil the water and produce steam. Calcium hydroxide is produced.

$$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(s)$$

In an *exothermic* reaction, *the products of the reaction have less (chemical) energy than the reactants*. In the reaction, *chemical energy* (stored in the bonds of chemicals) *is converted to heat energy*, which is released to the surroundings. The temperature of the reaction mixture and its surroundings *goes up*.

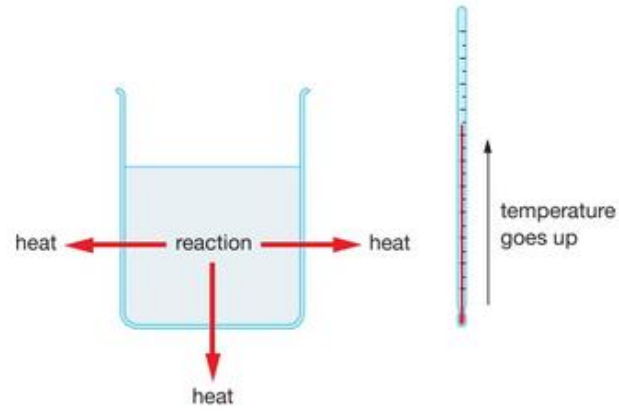
DID YOU KNOW?

Calcium oxide is known as **quicklime**. Adding water to it is described as *slaking* it. The calcium hydroxide produced is known as **slaked lime**.



▲ Figure 19.2 Calcium oxide reacting with water.

Safety Note: It is less hazardous to use a lump of calcium oxide, rather than powered calcium oxide



▲ Figure 19.3 In an exothermic reaction, chemical energy is converted to heat energy. Heat is released so the temperature goes up.

You will come across lots of exothermic reactions in this course. Some examples are given below.

COMBUSTION REACTIONS



▲ Figure 19.4 The burning of hydrogen is used in oxy-hydrogen cutting equipment underwater.

Any reaction that produces a flame is exothermic. Burning things produces heat energy.

For instance, hydrogen burns in oxygen, producing water and lots of heat:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

Apart from burning, other exothermic changes include:

- the reactions of metals with acids
- neutralisation reactions
- displacement reactions.

THE REACTIONS OF METALS WITH ACIDS

REMINDER

This reaction is described in detail in Chapter 14 (pages 152–153).

When magnesium reacts with dilute sulfuric acid, for example, the mixture gets very warm:

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

NEUTRALISATION REACTIONS

REMINDER

You can read about this reaction in Chapter 16 (page 170). We will investigate how much heat energy is released in a typical neutralisation reaction later in this chapter (pages 217–219).

About the only interesting thing that you can observe happening when sodium hydroxide solution reacts with dilute hydrochloric acid is that the temperature rises:

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

DISPLACEMENT REACTIONS

The thermite reaction between powdered aluminium and iron(III) oxide is a displacement (competition) reaction. This reaction releases a large amount of heat, which can be used in railway welding:

$$2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(l) + Al_2O_3(s)$$

ENTHALPY CHANGE OF A REACTION

KEY POINT

ΔH is pronounced 'delta H'. The Greek letter Δ is used to mean 'change in'. ΔH means 'change in heat'. Note: it is not possible to measure how much enthalpy (H) something has – you can only measure the change in enthalpy (ΔH) when it reacts.

REMINDER

The mole is a unit for the amount of a substance. You can read more about it in Chapter 5.

KEY POINT

The units for ΔH can be written as kJ/mol or kJ mol^{-1}.

You can measure the amount of heat energy taken in or released in a chemical reaction. It is called the **enthalpy change** of the reaction and is given the symbol ΔH. The enthalpy change is *the amount of heat energy taken in or given out in a chemical reaction*. It is the difference between the energy of the products and the energy of the reactants.

ΔH is given a minus or a plus sign to show whether heat is being given out or absorbed by the reaction. You always look at it from the point of view of the reactants. For an exothermic reaction, ΔH is given a *negative* number because the reactants are *losing* energy as heat. That heat is transferred to the surroundings, which then get warmer. ΔH is measured in units of kJ/mol (kilojoules per mole).

In an equation, the amount of heat given out or taken in can be shown as, for example:

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g) \quad \Delta H = -466.9\,kJ/mol$$

The ΔH written next to an equation represents the enthalpy change of the reaction, i.e. 466.9 kJ of heat is given out when *one mole* of magnesium reacts in this way. You know heat has been given out because ΔH has a negative sign.

CHEMISTRY ONLY

SHOWING AN EXOTHERMIC CHANGE ON AN ENERGY LEVEL DIAGRAM

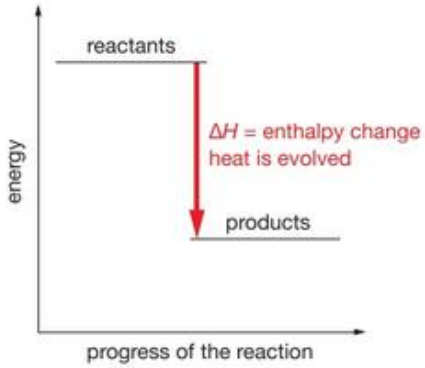
KEY POINT

The term **stability** is usually used to describe the relative energies of the reactants and the products in a chemical reaction. The more energy a chemical has, the less stable it is.

REMINDER

Remember that in a chemical reaction, the reactants are the chemicals you start with.

In an exothermic reaction, the reactants have more (chemical) energy than the products; we say that the products are *more stable* than the reactants. As the reaction happens, energy is given out in the form of heat. That energy warms up both the reaction itself and its surroundings.



▲ Figure 19.5 An exothermic change

END OF CHEMISTRY ONLY



▲ Figure 19.6 A boy using a cold pack to relieve pain in his elbow. Endothermic reactions occur in the chemical cold pack to absorb heat from the surroundings and reduce the temperature of his arm.

ENDOTHERMIC REACTIONS

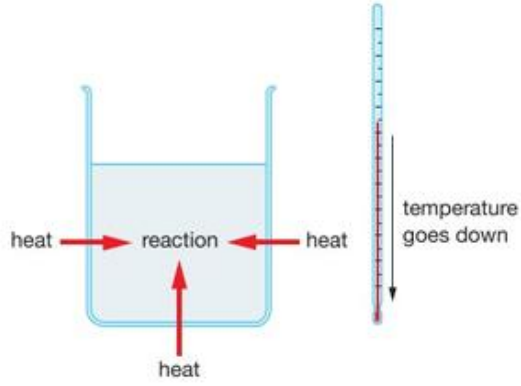
A reaction that absorbs heat from the surroundings is said to be **endothermic**. If you hold a test-tube in which an endothermic reaction is occurring you will notice that it *gets colder*.

In an endothermic reaction, *the products have more (chemical) energy than the reactants*. In order to supply the extra energy that is needed to convert the reactants (lower energy) into the products (higher energy), heat energy

REMINDER

Not all reactions that have to be heated are endothermic reactions. Sometimes we just heat something to make it go faster.

needs to be absorbed from the surroundings. This *heat energy is converted to chemical energy* (energy stored in the bonds of chemicals). The temperature of the reaction mixture and the surroundings *goes down* because heat energy has been converted into a different form of energy.



▲ Figure 19.7 In an endothermic reaction, heat energy is converted to chemical energy. Heat is absorbed so the temperature goes down.

You have seen the *thermal decomposition* of metal carbonates before in Chapter 13. These are examples of endothermic reactions. You have to heat a carbonate constantly to make it decompose.

For example, copper(II) carbonate (green) decomposes on heating to produce copper(II) oxide (black).

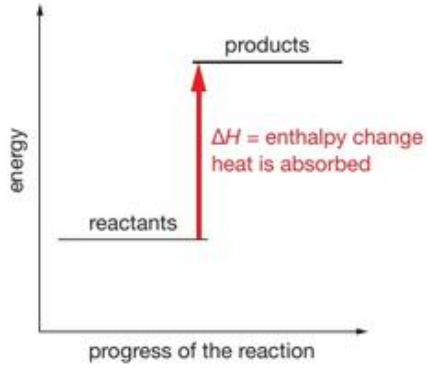
$$CuCO_3(s) \rightarrow CuO(s) + CO_2(g)$$

Similarly, zinc carbonate decomposes to form zinc oxide when heated.

$$ZnCO_3(s) \rightarrow ZnO(s) + CO_2(g)$$

CHEMISTRY ONLY

SHOWING AN ENDOTHERMIC CHANGE ON AN ENERGY LEVEL DIAGRAM



▲ Figure 19.8 An endothermic change

In an endothermic change, the products have more energy than the reactants so we say that the products are *less stable* than the reactants. That extra energy has to come from somewhere, and it is taken from the surroundings. In the case of the thermal decomposition of carbonates in the laboratory, it comes from the Bunsen burner.

Because the reactants are *gaining* energy, the enthalpy change of the reaction ΔH is given a *positive* sign.

For example:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g) \qquad \Delta H = +178 \text{ kJ/mol}$$

This means that 178 kJ of heat energy must be absorbed to convert 1 mole of calcium carbonate into calcium oxide and carbon dioxide.

END OF CHEMISTRY ONLY

MEASURING ENTHALPY CHANGES OF REACTIONS

Here we discuss how we can measure how much heat is taken in or given out by a chemical reaction, in other words the enthalpy change of a reaction.

SPECIFIC HEAT CAPACITY

KEY POINT

The unit for specific heat capacity is J/g/°C or J g^{-1} $°C^{-1}$.

When we heat something up, it gets hotter. The **specific heat capacity** tells us about *how much* energy has to be put in to increase the temperature of something. The specific heat capacity of a substance is defined as *the amount of heat needed to raise the temperature of 1 gram of a substance by 1 °C*.

For water, the value is 4.18 J/g/°C (joules per gram per degree Celsius). This means that 4.18 J of heat energy is needed if we want to increase the temperature of 1 g of water by 1 °C. If you want the temperature of 1 g of water to go up by 2 °C, then 4.18 × 2 = 8.36 J of heat energy must be supplied. If now you have 2 g of water, then 2 × 8.36 J of energy would be needed to raise the temperature by 2 °C.

KEY POINT

Again, we are using the delta (Δ) symbol to indicate a change, but this time a temperature change.
The concept of specific heat capacity also applies to the cooling of a substance. For example, to cool 1 g of water by 1 °C we need to take out 4.18 J of energy.

The amount of heat energy required is *directly proportional* to the mass (m) and the temperature change (ΔT) of the substance. The following equation can be used to calculate how much heat energy needs to be supplied to raise the temperature of mass m by ΔT °C:

heat energy change = mass × specific heat capacity × temperature change

$$Q = m \times c \times \Delta T$$

CALORIMETRY EXPERIMENTS FOR DETERMINING THE ENTHALPY CHANGES OF REACTIONS

It is fairly uncomplicated to measure the amount of heat absorbed or given out in several kinds of chemical reactions and physical changes. The technique used to do this is called **calorimetry** and it is based on the idea that if we use the heat from a reaction to heat another substance, such as water, we can then use the equation introduced above ($Q = mc\Delta T$) to calculate the amount of heat released. Here the mass, the specific heat capacity and the temperature change are all referring to *the substance heated*. If we know how many moles of reactants are used in the reaction, we can then work out the **molar enthalpy change**, ΔH, of the reaction in the unit kJ/mol.

The following activity illustrates how we can use calorimetry to determine the molar enthalpy change of combustion of an alcohol, i.e. how much heat energy is released when 1 mole of alcohol burns.

ACTIVITY 1

▼ PRACTICAL: MEASURING ENTHALPY CHANGES IN COMBUSTION REACTIONS

!

Safety Note: Wear eye protection. Do not carry a lit spirit burner and do not fill or re-fill one when there is a naked flame nearby.

One of the most common calorimetry experiments at International GCSE is to measure the amount of heat given off when a number of small alcohols are burned. You could use methanol, ethanol, propan-1-ol and butan-1-ol (see Chapter 22 for an introduction to the naming of alcohols).

The alcohols are burned in a small spirit burner, and the heat produced is used to heat some water in a copper can (the calorimeter).

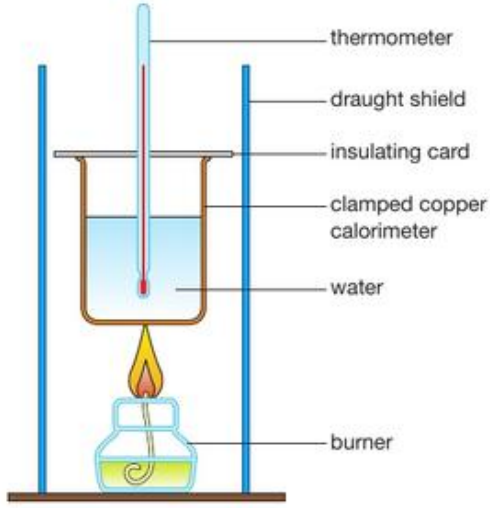
KEY POINT

A *calorimeter* is simply something that we do a calorimetry experiment in; here it is a copper can, in other experiments it can be a polystyrene cup.

The following procedure could be used:

- Measure 100 cm^3 of cold water using a measuring cylinder and transfer the water to a copper can.
- Take the initial temperature of the water.
- Weigh a spirit-burner containing ethanol with its lid *on*. The lid should be kept on when the wick is not lit to prevent the alcohol from evaporating.

- Arrange the apparatus as shown in Figure 19.9 so that the spirit-burner can be used to heat the water in the copper can. The apparatus is shielded as far as possible to prevent draughts.
- Light the wick to heat the water. Stop heating when you have a reasonable temperature rise of water (say, about 40.0 °C). The flame can be extinguished by putting the lid back on the wick.
- Stir the water thoroughly and measure the maximum temperature of the water.
- Weigh the spirit-burner again with its lid *on*.
- The experiment can be repeated with the same alcohol to check for reliability, and then carried out again with whatever other alcohols are available.



▲ Figure 19.9 A calorimetry experiment to measure the enthalpy change of combustion of alcohols.

KEY POINT

You may see other versions of this experiment recommending a different temperature rise or suggesting that you heat the water for a fixed number of minutes. There's no ideal answer to this. The higher the temperature the water reaches, the greater the heat losses during the experiment. On the other hand, if you just increase the temperature of the water by a small amount, errors in reading the thermometer or in finding the mass change of the alcohol become too significant.

SAMPLE DATA

volume of water/cm^3	100
mass of burner + ethanol before experiment/g	137.36
mass of burner + ethanol after experiment/g	136.58
original temperature of water/°C	21.5
final temperature of water/°C	62.8

Combustion is an exothermic reaction so the temperature of the water goes up. As ethanol is burned, the total mass of the burner and ethanol goes down.

Using the above data we can determine how much heat is released when one mole of an alcohol is burned, that is, the molar enthalpy change of combustion.

CALCULATIONS FOR ACTIVITY 1

We are going to use the equation $Q = mc\Delta T$ so we need to find out what each quantity is.

Temperature change of water = ΔT = 62.8 – 21.5 = 41.3 °C.

Mass of water being heated = m = 100 g, the density of water is approximately 1 g/cm^3 at room temperature, so 100 cm^3 of water has a mass of 100 g.

c is the specific heat capacity of the water (it is the water that is being heated): $c = 4.18\,J/g/°C$

Heat gained by water = $Q = mc\Delta T = 100 \times 4.18 \times 41.3 = 17\,260\,J$.

Divide Q by 1000 to give energy in kJ = 17.26 kJ.

KEY POINT

The enthalpy change when 1 mole of a substance is burned is called the *enthalpy change of combustion*.

This means 17.26 kJ of heat energy is released by the combustion of the ethanol in this experiment. To calculate the amount of heat produced when 1 mole of ethanol, CH_3CH_2OH, burns, you need to find out how many moles of ethanol are burned in your experiment.

Mass of ethanol burned = decrease in the mass of the burner
= 137.36 – 136.58 = 0.78 g

HINT

We have quoted the final answer to *3 significant figures*. In the exam, it is usually best to quote answers to 3 significant figures. When we are doing experiments we should actually base our answers on *the piece of data with the fewest significant figures*. When you are doing calculations carry more significant figures through on your calculator and then round the value at the end.

Ethanol has the formula C_2H_5OH. We can calculate the relative molecular mass of ethanol by adding up the relative atomic masses:
$(2 \times 12) + (5 \times 1) + 16 + 1 = 46$

Number of moles (n) of ethanol, CH_3CH_2OH, burned

$$n = \frac{\text{mass } (m)}{\text{relative molecular mass } (M_r)} = \frac{0.78}{46} = 0.01696\,\text{mol}$$

Molar enthalpy change of combustion of ethanol (ΔH)

$$\Delta H = \frac{\text{heat energy change } (Q)}{\text{number of moles of ethanol burned } (n)}$$

$$= \frac{17.26}{0.01696} = 1020\,\text{kJ/mol}$$

HINT

Always remember to put in the sign of the enthalpy change at this stage. You will be able to see whether the reaction is endothermic (ΔH +) or exothermic (ΔH –) from looking at the original temperature change: does it go up (exothermic) or down (endothermic)?

The amount of heat released in the complete combustion of 1 mole of ethanol is therefore:

$$CH_3CH_2OH + 3O_2 \rightarrow 2CO_2 + 3H_2O \qquad \Delta H = -1020\ \text{kJ/mol}$$

The negative sign shows that heat is released and the combustion reaction is exothermic.

EVALUATION OF THE EXPERIMENTAL RESULTS

How accurate is the figure of –1020 kJ/mol for the enthalpy change of the combustion of ethanol? The accepted value found in data booklets for ethanol is –1370 kJ/mol, which means that 1370 kJ of heat should be given out when 1 mole of ethanol burns. You can see that the value we obtained is less exothermic than expected; our reaction seemed to give out less heat than expected.

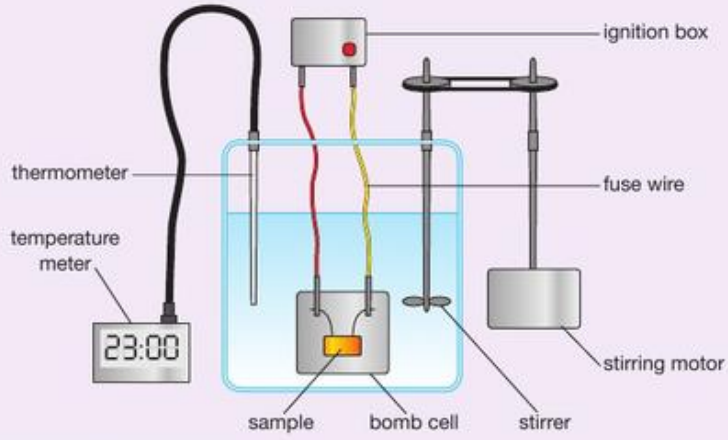
There are many sources of error in this experiment, in particular *large amounts of heat loss*.

The warm water gives out heat to the air, heat is lost from the flame, which goes straight into the air rather than into the water, and heat is used to raise the temperature of the copper can and the thermometer.

Another major source of error is the *incomplete combustion* of alcohol. We will talk about incomplete combustion in more detail in Chapter 23. Incomplete combustion of an alcohol occurs when there is not enough oxygen present. Incomplete combustion releases less heat than complete combustion. We can see that the combustion in Activity 1 is incomplete because the flame of the wick is often yellow orange rather than blue, and soot (carbon) is produced at the bottom of the copper can. If the combustion is complete, the flame should be blue and carbon dioxide should be produced instead of carbon.

EXTENSION WORK

A bomb calorimeter can be used to obtain more accurate values of the enthalpy changes of combustion. The substance to be combusted is ignited electrically and burns in the presence of pure oxygen. The heat transferred to both water and the calorimeter itself is determined to give a more accurate value of the enthalpy change.



▲ Figure 19.10 Bomb calorimeter

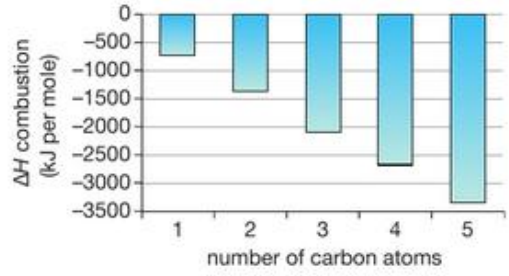
That doesn't mean you can't use this experiment to make useful comparisons. If you repeat it with other alcohols, under conditions that are as similar as possible, you can find how the heat evolved (given out) changes as the alcohol gets bigger.

In Figure 19.11 we can see that the combustion reaction gets more exothermic as the alcohol chain becomes longer. In other words, longer alcohols give out more heat energy per mole when they burn than shorter ones.

The increase in the heat evolved as alcohol chain lengthens is regular. The difference between one alcohol and the next is always an extra CH_2 (see Chapter 26, page 288, for more detail on the structures of alcohols), and so the number of extra bonds broken and made (see below) increases in a regular way. That means that the heat evolved will also increase in a regular way.

KEY POINT

You may sometimes see these plotted as smooth graphs, but that is technically wrong. A smooth curve should only be used for a continuous independent variable, one which can take any value. There is no such thing as an alcohol with 0.5 or 1.64 carbon atoms! The number of carbon atoms is a **non-continuous variable** because it can only take whole number values.



▲ Figure 19.11 Molar enthalpy change of the combustion of alcohols.

WORKING OUT ENTHALPY CHANGES FOR REACTIONS INVOLVING SOLUTIONS USING CALORIMETRY EXPERIMENTS

You can use very similar methods for measuring molar enthalpy changes in displacement reactions (e.g. zinc and copper(II) sulfate solution), dissolving (e.g. dissolving ammonium chloride in water to form a solution) and neutralisation reactions (e.g. between potassium hydroxide solution and dilute hydrochloric acid). These experiments also involve heating water, but this time the water is part of the solutions we are using. We will look at how to do this in the next few practicals.

ACTIVITY 2

▼ PRACTICAL: MEASURING ENTHALPY CHANGES FOR DISPLACEMENT REACTIONS

! Safety Note: Avoid skin contact with the chemicals.

In order to determine the enthalpy change of the reaction of zinc and copper(II) sulfate, the following procedure could be used:

- Place a polystyrene cup in a 250 cm^3 glass beaker.
- Transfer 50 cm^3 of 0.200 mol/dm^3 copper(II) sulfate solution into the polystyrene cup using a measuring cylinder.
- Weigh 1.20 g of zinc using a weighing boat on a balance.
- Record the initial temperature of the copper(II) sulfate solution.
- Add the zinc.
- Stir the solution as quickly as possible.
- Record the maximum temperature reached.



▲ 19.12 A calorimetry experiment to measure the enthalpy change of a displacement reaction.

SAMPLE DATA

initial temperature of copper(II) sulfate solution/°C	17.0
maximum temperature of the reaction mixture/°C	27.3

We can use this data to calculate the enthalpy change for this displacement reaction, when 1 mole of copper(II) sulfate reacts with zinc.

CALCULATIONS FOR ACTIVITY 2

Heat given out in this reaction: $Q = mc\Delta T = 50 \times 4.18 \times (27.3 - 17.0)$

$= 2152.7\,J$

$= 2.1527\,kJ$

Here we assume the following:

1. The density of the copper sulfate solution is the same as that of water, so 1 cm^3 of solution has a mass of 1 g.
2. The specific heat capacity of the mixture is the same as that of water. This is a fairly reasonable assumption because the reaction mixture is mostly water.

In this experiment we have used excess zinc. *Excess* means more than enough zinc is present to ensure all the copper(II) sulfate reacts. If you calculate the number of moles of copper(II) sulfate and the number of moles of zinc used in this procedure, you should spot that the number of moles of zinc used is more than that of copper(II) sulfate:

$$\text{number of moles } (n) \text{ of zinc added} = \frac{\text{mass } (m)}{\text{relative atomic mass } (A_r)}$$

$$= \frac{1.20}{65}$$

$$= 0.0185 \text{ mol}$$

$$\text{number of moles } (n) \text{ of copper(II) sulfate added} = \text{volume } (V) \times \text{concentration } (C)$$

$$= 0.050 \times 0.200$$

$$= 0.0100 \text{ mol}$$

Now we need to calculate how much heat is released when 1 mole of copper sulfate reacts with excess zinc:

Molar enthalpy change of reaction (ΔH)

$$\Delta H = \frac{\text{heat energy change } (Q)}{\text{number of moles of copper sulfate reacted } (n)}$$

$$= \frac{2.1527}{0.0100} = 215 \text{ kJ/mol}$$

The amount of heat released in the displacement reaction when 1 mole of $CuSO_4$ reacts with excess Zn is therefore:

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s) \qquad \Delta H = -215 \text{ kJ/mol}$$

We have added the negative sign because we know that the temperature of the reaction mixture went up. The negative sign shows that this is an exothermic reaction; heat is released.

KEY POINT

1 mol of Zn reacts with 1 mol of $CuSO_4$, therefore 0.0100 mol of $CuSO_4$ reacts with 0.0100 mol of Zn. We added 0.0185 mol of Zn, so Zn is in excess.

REMINDER

Remember that we have to divide the volume by 1000 to convert it to dm^3 because the concentration is given in mol/dm^3.

EXTENSION WORK

You could repeat this experiment with metals of different reactivities. The more reactive a metal is, the more heat should be released in the displacement reaction. Make sure you keep everything else the same: the number of moles of the metals, the size of the solid particles, and the volume and concentration of the copper(II) sulfate solution. Do not use metals that are more reactive than magnesium, otherwise you are measuring the heat released when the metal reacts with water instead!

!

Safety Note: Wear eye protection and avoid skin contact with the salts and their solutions.

ACTIVITY 3

▼ PRACTICAL: MEASURING ENTHALPY CHANGES WHEN SALTS DISSOLVE IN WATER

We can also use calorimetry experiments to work out the amount of heat given out/taken in when salts dissolve in water. The following procedure could be used:

- Place a polystyrene cup in a 250 cm^3 glass beaker.
- Transfer 100 cm^3 of water into the polystyrene cup using a measuring cylinder.
- Record the initial temperature of the water.
- Weigh 5.20 g of ammonium chloride using a weighing boat on a balance.
- Add the ammonium chloride to water and stir the solution vigorously until all the ammonium chloride has dissolved.
- Record the minimum temperature.

The set-up is very similar to the one used in Activity 2, see Figure 19.12.

SAMPLE DATA

initial temperature of water/°C	18.3
minimum temperature of salt solution/°C	15.1

Note: the temperature of the water decreases, so the reaction is endothermic and heat is absorbed from the surroundings for the dissolving process to occur.

We can use this data to calculate the molar enthalpy change for dissolving ammonium chloride.

CALCULATIONS FOR ACTIVITY 3

Heat absorbed: $Q = mc\Delta T = 100 \times 4.18 \times (18.3 - 15.1) = 1337.6\,\text{J} = 1.3376\,\text{kJ}$

Here we assume the following:

1. The specific heat capacity of the diluted solution of ammonium chloride is the same as that of water. This is quite a reasonable assumption because the mixture is mostly water.
2. The mass of the solution is 100 g. The mass of the ammonium chloride is relatively small and it is ignored in the calculation. There are other major sources of error in the experiment, for example heat absorbed from the surrounding air, which makes much more difference to the results.

Now we need to calculate how much heat is absorbed when 1 mole of ammonium chloride dissolves in excess water.

The relative formula mass of ammonium chloride (NH_4Cl) is $14 + 4 \times 1 + 35.5 = 53.5$.

Number of moles (n) of ammonium chloride dissolved

$$n = \frac{\text{mass } (m)}{\text{relative formula mass } (M_r)} = \frac{5.20}{53.5} = 0.0972\,\text{mol}$$

Molar enthalpy change of solution (ΔH)

$$\Delta H = \frac{\text{heat energy change } (Q)}{\text{number of moles of ammonium chloride dissolved } (n)}$$

$$= \frac{1.3376}{0.0972}$$

$$= 13.8\,\text{kJ/mol}$$

KEY POINT

Dissolving is a physical process rather than a chemical reaction. When some salts dissolve in water heat is given out (exothermic), but when others dissolve the process is endothermic.

The amount of heat absorbed when dissolving 1 mole of ammonium chloride in water is therefore:

$$NH_4Cl(s) \rightarrow NH_4Cl(aq) \qquad \Delta H = +13.8\,\text{kJ/mol}$$

The positive sign shows that heat is absorbed and the dissolving of ammonium chloride in water is an endothermic process (the temperature goes down).

ACTIVITY 4

▼ PRACTICAL: MEASURING ENTHALPY CHANGES OF NEUTRALISATION BETWEEN AN ALKALI AND AN ACID

! Safety Note: Wear eye protection and avoid skin contact with the acid and alkali.

The reaction between an alkali and an acid is essentially between OH^- and H^+ ions to form water (see Chapter 16):

$$OH^-(aq) + H^+(aq) \rightarrow H_2O(l)$$

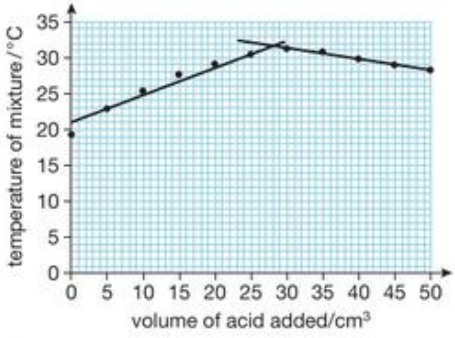
Let's look at the reaction between potassium hydroxide (an alkali) and hydrochloric acid. Suppose you know the concentration of the

potassium hydroxide but are not so sure of the concentration of the dilute hydrochloric acid solution (the label has gone missing!). The following method could be used to find out the concentration of the acid and how much heat is released during the neutralisation reaction.

- Place a polystyrene cup in a 250 cm^3 glass beaker.
- Transfer 25 cm^3 of 2.00 mol/dm^3 potassium hydroxide into the polystyrene cup using a measuring cylinder.
- Record the initial temperature.
- Fill a burette with 50.00 cm^3 of dilute hydrochloric acid.
- Use the burette to add 5.00 cm^3 of dilute hydrochloric acid to the potassium hydroxide.
- Stir vigorously and record the maximum temperature reached.
- Continue adding further 5.00 cm^3 portions of dilute hydrochloric acid to the cup, stirring and recording the maximum temperature each time, until a total volume of 50.00 cm^3 has been added.

We can plot a graph of the temperature of the mixture versus the volume of acid added.

SAMPLE DATA



▲ Figure 19.13 Temperature change in the neutralisation reaction between KOH and HCl.

In this reaction the temperature increases at first but then decreases. The reaction between the acid and the alkali is exothermic. At the beginning the temperature goes up because the acid reacts with the alkali, giving out heat. But when all the alkali has been used up, we are just adding cold acid to our warm solution (there is no reaction because there is nothing for the acid to react with) and the temperature goes down.

Two lines of best fit can be drawn on our graph. The point where the lines cross represents complete neutralisation. From the graph, we can identify the maximum temperature reached during the experiment as 31.8 °C. At this point, 28.00 cm^3 of acid has been used.

CALCULATIONS FOR ACTIVITY 4

To work out how much heat is released in the neutralisation reaction, we can use the same calculation as before.

The total volume of the solution at the neutralisation point is 25 + 28 = 53 cm^3.

Heat given out is shown by $Q = mc\Delta T = 53 \times 4.18 \times (31.8 - 19.3) = 2769.3$ J.

If we divide by 1000 we get 2.7693 kJ.

Here we assume the following:

1. The density of the reaction mixture is the same as that of water, so 1 cm^3 of solution has a mass of 1 g.
2. The specific heat capacity of the mixture is the same as that of water. This is a fairly reasonable assumption because the neutralised solution is mostly water.

Now we need to calculate how much heat is released when 1 mole of potassium hydroxide reacts with 1 mole of hydrochloric acid:

$$\text{Number of moles of KOH } (n) = \text{concentration } (C) \times \text{volume } (V)$$

$$= 2.00 \times \frac{25}{1000} = 0.0500 \text{ mol}$$

The number of moles of HCl will be exactly the same because we have taken the temperature at the neutralisation point.

Molar enthalpy change of neutralisation (ΔH)

$$\Delta H = \frac{\text{heat energy change } (Q)}{\text{number of moles of KOH or HCl reacted } (n)}$$

$$= \frac{2.7693}{0.0500} = 55.4\,\text{kJ/mol}$$

The amount of heat released in the neutralisation reaction when 1 mole of KOH reacts with 1 mole of HCl is therefore:

$$KOH(aq) + HCl(aq) \rightarrow KCl(aq) + H_2O(l) \qquad \Delta H = -55.4\,\text{kJ/mol}$$

The negative sign shows that heat is released in this exothermic neutralisation reaction.

CHEMISTRY ONLY

We can also work out the unknown HCl concentration.

We know that there was 0.0500 mol KOH originally present.

28.00 cm^3 of HCl was required for neutralisation therefore this contained 0.0500 mol of HCl:

$$\text{Concentration of HCl } (C) = \frac{\text{number of moles of HCl } (n)}{\text{volume of HCl used for neutralisation } (V)}$$

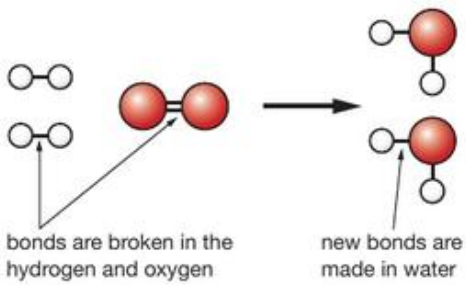
$$= \frac{0.0500}{0.02800} = 1.79\,\text{mol/dm}^3$$

WHY DO REACTIONS EITHER GIVE OUT OR ABSORB HEAT?

During chemical reactions bonds in the reactants have to be broken and new ones formed to make the products. *Breaking bonds needs energy (endothermic) and making bonds releases energy (exothermic).*

Think about what happens when hydrogen burns in oxygen to make water:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$



▲ Figure 19.14 Hydrogen burning in oxygen to make water

KEY POINT

If you want to be really accurate about this, the water would originally be formed as steam. Heat is also given out when the steam condenses to form liquid water. This is due to formation of *intermolecular forces* of attraction between the molecules in liquid water. Formation of attractive forces between molecules is also exothermic.

Energy has to be *supplied* to *break the bonds* in the hydrogen molecules and in the oxygen molecule. Energy is *released* when *new bonds are formed* between the hydrogen and oxygen atoms in the water molecules.

In this particular reaction, the products are more stable than the reactants. This means the total amount of energy released when covalent bonds are formed in the product (water molecules) is more than is required to break the covalent bonds in the reactants (hydrogen and oxygen molecules). The reaction is exothermic.

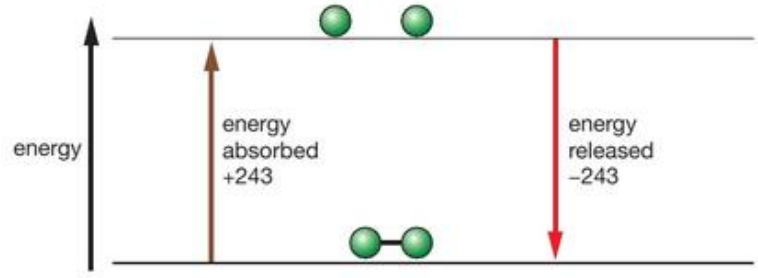
When you heat calcium carbonate, breaking up the original bonds in the compound uses more energy than you obtain when the new ones are made.

The reactants are more stable than the products. That means that when the reaction is finished, more energy has been absorbed than is released, an endothermic change.

CALCULATION OF ENTHALPY CHANGES OF REACTION USING BOND ENERGIES

BOND ENERGIES

Breaking chemical bonds requires energy. The stronger the bond, the more energy is needed to break it. Bond energies are measured in kJ/mol (kilojoules per mole). For example, the Cl–Cl bond energy is 243 kJ/mol. This means that it will take 243 kJ to break all the Cl–Cl bonds in *1 mole* of chlorine *gas*. The bond energy also represents the amount of energy released when 1 mole of the bonds form. Figure 19.15 shows that it needs an input of 243 kJ of energy to break 1 mole of chlorine molecules into atoms. If the atoms recombine into their original molecules, then exactly the same amount of energy will be released again.



▲ Figure 19.15 Breaking and making bonds between chlorine atoms

- *Breaking bonds needs energy: endothermic.*
- *Making bonds releases energy: exothermic.*

KEY POINT

Bond energy is defined as *the amount of energy needed to break 1 mole of covalent bonds in gaseous molecules.*

The H–Cl bond energy is 432 kJ/mol. If you see '2HCl(g)' in an equation, it will take 2 × 432 kJ to break all the H–Cl bonds in the two moles of hydrogen chloride gas. Or, 2 × 432 kJ will be released when 2HCl(g) is formed from H and Cl atoms.

Table 19.1 The energy needed to break bonds

Bond	C–H	C–Cl	C–I	Cl–Cl	I–I	H–Cl	H–I
Bond energy/kJ/mol	413	346	234	243	151	432	298

You can see that some bonds are much stronger than others, for example the bond between iodine and hydrogen is about twice as strong as the bond between two iodine atoms.

Bond energies can be used to calculate how much heat will be absorbed or released during reactions of *covalent* molecules. If a reaction involves *ionic* compounds, different energy terms have to be used, which are beyond International GCSE.

CALCULATION OF THE HEAT RELEASED OR ABSORBED DURING A REACTION

You can estimate the heat energy released or absorbed in a reaction by calculating how much energy would be needed to break the substances up into individual atoms, and then how much would be given out when those atoms recombine into new arrangements.

EXTENSION WORK

Bond energy calculations only ever give *estimates* of the amount of heat evolved or absorbed. They never give exact values. The strength of a bond varies slightly depending on what is around it in the molecule. Quoted bond energies are usually *average* values, which give fairly good, but not perfect, answers for enthalpy changes of reactions. The enthalpy changes of reactions measured using calorimetry experiments are more specific for particular compounds.

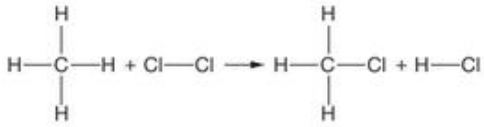
For example:

if:	energy needed to break all the bonds in reactants	=	+1000 kJ (positive sign as the process if endothermic)
and:	energy released when new bonds are made in products	=	−1200 kJ (negative sign as the process is exothermic)
then:	overall change	=	−200 kJ (the overall reaction is exothermic)

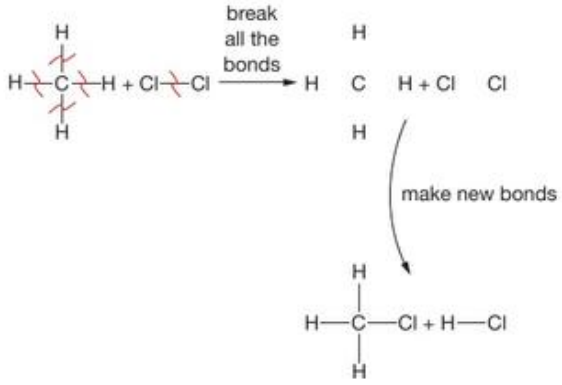
This reaction would release 200 kJ of energy.

THE REACTION BETWEEN METHANE AND CHLORINE

$$CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$$



Methane reacts with chlorine in the presence of ultraviolet light to produce chloromethane and hydrogen chloride (see Chapter 24, page 280, for more details on this reaction). You can imagine all the bonds being broken in the methane and chlorine, and then being reformed in new ways in the products:



EXTENSION WORK

The real reaction doesn't actually happen by all the bonds being broken in this way. That isn't important. The overall amount of heat released or absorbed is the same (as long as the reactants and the products are exactly the same) however you do the reaction, even if one of the routes you use to calculate it is entirely imaginary! This is summarised in an important law called *Hess's law*, which you will meet if you do chemistry at a higher level.

You can calculate the energy needed to break all the bonds and the energy given out as new ones are made.

Bonds that need to be broken (endothermic):

4 C–H bonds	= 4 × (+413)	=	+ 1652 kJ
1 Cl–Cl bond	= 1 × (+243)	=	+ 243 kJ
total		=	+ 1895 kJ

New bonds made (exothermic):

3 C–H bonds	= 3 × (–413)	=	–1239 kJ
1 C–Cl bond	= 1 × (–346)	=	–346 kJ
1 H–Cl bond	= 1 × (–432)	=	–432 kJ
total		=	–2017 kJ

HINT

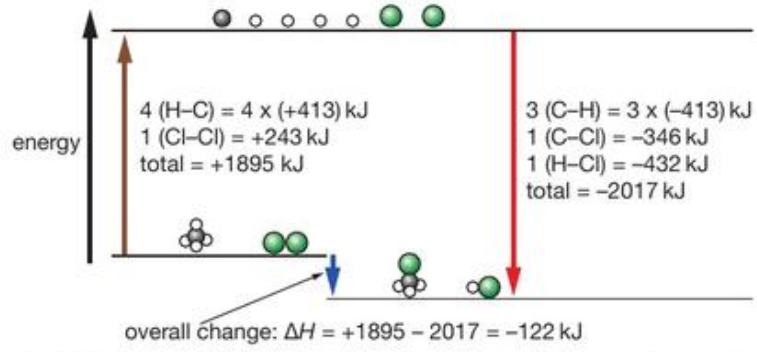
Remember that when energy is given out, you show this by putting a negative sign in front of the value.

We can see that more energy is released when bonds are formed than is required to break them, so the reaction releases energy overall; it is exothermic.

The overall energy change is +1895 + (–2017) kJ = –122 kJ.

The negative sign of the answer shows that, overall, heat is given out as the bonds rearrange.

You can show all this happening on an energy level diagram, as in Figure 19.16.



▲ Figure 19.16 Methane reacts with chlorine to produce chloromethane and hydrogen chloride.

AN EXAMPLE INVOLVING DOUBLE BONDS

The bond energy for the O=O double bond is 498 kJ/mol. This is the energy required to break the whole of the double bond.

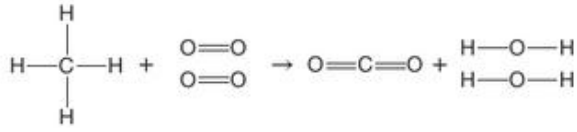
The equation for the combustion of methane is:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

Calculate the enthalpy change for this reaction given the bond energies in the table and draw an energy level diagram for the reaction.

Bond	C–H	O=O	C=O	H–O
Bond energy/kJ/mol	413	498	743	464

We start by drawing out the structures showing all the bonds so that you can make sure that we get the number of bonds correct:



Bonds that need to be broken (endothermic):

4 C–H bonds	= 4 × (+413)	=	+1652 kJ
2 O=O bonds	= 2 × (+498)	=	+996 kJ
total		=	+2648 kJ

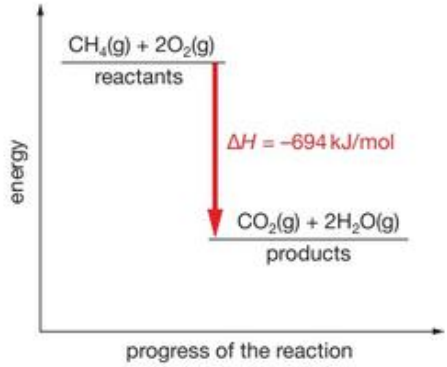
New bonds made (exothermic):

2 C=O bonds	= 2 × (–743)	=	–1486 kJ
4 O–H bonds	= 4 × (–464)	=	–1856 kJ
total		=	–3342 kJ

Enthalpy change = +2648 + (–3342) = –694 kJ/mol.

More energy is released when bonds are formed than is required to break them; the reaction is exothermic.

Energy is given out when the reactants are converted to the products. Therefore we know that the products must have less energy than the reactants; the products are more stable. We can show this on an energy level diagram (Figure 19.17).



▲ Figure 19.17 An energy level diagram for the combustion of methane

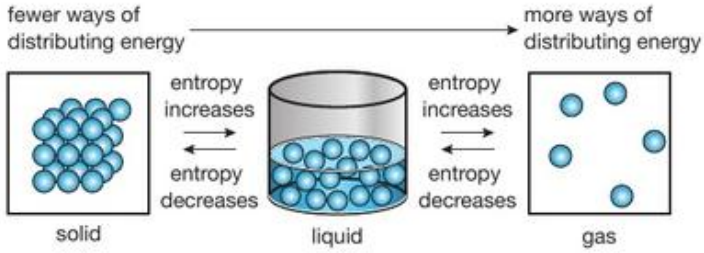
EXTENSION WORK

The reaction has been shown with $H_2O(g)$ rather than $H_2O(l)$ because bond energies can only be used for gaseous molecules. We could find out the enthalpy change for this reaction forming $H_2O(l)$ if we knew the amount of energy released in the process $H_2O(g) \rightarrow H_2O(l)$. This is beyond International GCSE level.

END OF CHEMISTRY ONLY

LOOKING AHEAD – ENTROPY

Many reactions we meet at International GSCE level are exothermic, for example combustion and neutralisation. Despite requiring a small amount of energy to get the reactions started (*activation energy*), these reactions release heat energy overall because the products are more stable than the reactants. This shows that enthalpy is certainly a driving force for a reaction to occur. However, endothermic processes do occur. How do they happen? Dissolving ammonium nitrate into water causes the temperature of the solution to decrease, but it dissolves anyway. This is possible because of a different factor called entropy. *Entropy is a measure of the number of ways for the energy in a system to be distributed*. It is greater in a more disordered system because the energy can be dispersed more flexibly. It follows that solids have lower entropy than liquids, which have lower entropy than gases, as shown in Figure 19.18.



▲ Figure 19.18 Entropy is greater in a more disordered system.

If the entropy of a system increases during a reaction, it is more likely for that reaction to occur. When dissolving ammonium nitrate in water, although the process is endothermic, the NH_4^+ and NO_3^- ion particles in the giant ionic lattice become separated from each other. The ions have more freedom of movement in water and the entropy of the system increases, which outweighs the effect of enthalpy.

The reaction between baking soda and vinegar is another example of an endothermic reaction that occurs spontaneously. The reaction produces a gas which leads to an increase in the entropy of the system.



▲ Figure 19.19 The reaction between baking soda and vinegar is used to make papier mâché volcanos.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING, PROBLEM SOLVING (9)

1 a Explain what is meant by an *exothermic reaction* and write balanced chemical equations for any two exothermic changes (apart from the combustion of heptane given in part b).

CHEMISTRY ONLY

SKILLS INTERPRETATION (10)

b Heptane, C_7H_{16}, is a hydrocarbon found in petrol. The equation for the combustion of heptane is:

$$C_7H_{16}(l) + 11O_2(g) \rightarrow 7CO_2(g) + 8H_2O(l) \qquad \Delta H = -4817\,kJ/mol$$

Draw an energy level diagram to show the combustion of heptane. Label the axes and show clearly the reactants, the products and the enthalpy change of reaction on your diagram.

SKILLS REASONING (9)

c Explain in terms of breaking and making bonds why the reaction in b is exothermic.

END OF CHEMISTRY ONLY

(7)

2 a Explain what is meant by an *endothermic reaction*.

CHEMISTRY ONLY

SKILLS INTERPRETATION (10)

b Nitrogen and oxygen gases can combine to form nitrogen monoxide as follows:

$$N_2(g) + O_2(g) \rightarrow 2NO(g) \qquad \Delta H = +180\,kJ/mol$$

Draw an energy level diagram for this process. Label the axes and show clearly the reactants, the products and the enthalpy change of reaction on your diagram.

END OF CHEMISTRY ONLY

SKILLS CRITICAL THINKING (9)

3 Identify each of the following changes as exothermic or endothermic. In some cases you will have to rely on your previous knowledge of chemistry. Several reactions are likely to be completely new to you.

a The reaction between sodium and water.

b Burning methane (major constituent of natural gas).

c The reaction between sodium carbonate and ethanoic acid. A thermometer placed in the reaction mixture shows a temperature drop.

d $S(s) + O_2(g) \rightarrow SO_2(g) \qquad \Delta H = -297\,kJ/mol$

e $CuSO_4(s) \rightarrow CuO(s) + SO_3(g) \qquad \Delta H = +220\,kJ/mol$

f If you dissolve solid sodium hydroxide in water, the solution becomes very hot.

SKILLS INTRAPERSONAL (11)

4 Self-heating cans are used to provide warm food in situations where it is inconvenient to use a more conventional form of heat. By doing an internet (or other) search, find out how self-heating cans work. Write a short explanation of your findings (not exceeding 200 words). You should include equation(s) for any reaction(s) involved, and a diagram or picture if it is useful.

CHEMISTRY ONLY

SKILLS REASONING, PROBLEM SOLVING

10

5 Use the bond energies in the table to calculate the amount of heat released or absorbed when the following reactions take place. In each case, say whether the change is exothermic or endothermic.

Bond	Bond energy/kJ/mol
C–H	413
C–Br	290
Br–Br	193
H–Br	366
H–H	436
Cl–Cl	243
H–Cl	432
O=O	498
O–H	464
N≡N	944
N–H	388

a $CH_4(g) + Br_2(g) \rightarrow CH_3Br(g) + HBr(g)$

(The structure of CH_3Br is the same as that of CH_3Cl)

b $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$

c $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$

d $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

END OF CHEMISTRY ONLY

SKILLS EXECUTIVE FUNCTION

4

6 A student investigated the amount of heat given out when hexane, C_6H_{14}, burns using the apparatus in Figure 19.9. Hexane is a highly flammable liquid which is one of the components of petrol (gasoline).

In each case, the student calculated the amount of heat evolved per mole of hexane. Her first two experiments produced answers of –3200 and –3900 kJ/mol of heat evolved. She then decided to do a third experiment. Her results were as follows:

volume of water in copper calorimeter/cm^3	100
mass of burner + hexane before experiment/g	35.62
mass of burner + hexane after experiment/g	35.23
original temperature of water/°C	19.0
final temperature of water/°C	55.0

a Suggest a reason why the student decided to do a third experiment.

5

b Apart from wearing eye protection, suggest two other safety precautions the student should take during the experiment.

SKILLS PROBLEM SOLVING (9)

c Use the results table to calculate the amount of heat in **kJ** evolved by the burning hexane during the experiment.

(Specific heat capacity of water = 4.18 J/g/°C; density of water = 1 g/cm^3.)

d Calculate the amount of heat energy released per gram of hexane.

(10) e Calculate the molar enthalpy change of combustion of hexane. (A_r: H = 1, C = 12)

SKILLS REASONING (8) f To calculate the average value of the heat evolved per mole when hexane burns, the student took an average of her results. She decided not to use the figure of –3900 in calculating the average because it was so different from the other two. Suggest two reasons why this reaction might have been more exothermic than the others.

(9) g A data book gave a figure of –4194 kJ of heat evolved when 1 mole of hexane burns. Suggest two reasons why all the results in the student's experiment were less negative than this.

SKILLS PROBLEM SOLVING

7 When 5.15 g of lithium chloride (LiCl) is dissolved in 50 cm^3 of water the temperature of the solution goes from 17.0 °C to 33.5 °C.

a Calculate the heat energy released in this experiment. (Specific heat capacity of water = 4.18 J/g/°C; mass of 1 cm^3 of solution = 1 g.)

b Calculate the amount, in moles, of lithium chloride dissolved in the solution.

(10) c Using your answers from a and b, calculate the enthalpy change when 1 mol of lithium chloride dissolves in water in kJ/mol.

SKILLS REASONING (9) d A data book gives a figure of –37.2 kJ/mol for heat released when 1 mole of lithium chloride is dissolved in water. Suggest two reasons why the value you calculated in c might be different from the value given in the data book.

CHEMISTRY ONLY

SKILLS PROBLEM SOLVING (10)

8 To completely neutralise 200 cm^3 of 0.500 mol/dm^3 sodium hydroxide (NaOH), a student adds 100 cm^3 of 0.500 mol/dm^3 sulfuric acid (H_2SO_4). The temperature of the solution goes up by 4.50 °C.

The equation for the reaction is

$$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

a Calculate the amount, in moles, of NaOH in the sodium hydroxide solution.

b Calculate the heat released in this reaction. (Specific heat capacity of solution = 4.18 J/g/°C; mass of 1 cm^3 of solution = 1 g.)

c Using your answers from a and b, calculate the enthalpy change when 1 mol of sodium hydroxide is neutralised by sulfuric acid in kJ/mol.

SKILLS INTERPRETATION d Draw an energy level diagram for this reaction. Label the axes and show clearly the reactants, the products and the enthalpy change of reaction on your diagram.

SKILLS PROBLEM SOLVING e Predict the temperature rise if the experiment were repeated using 200 cm^3 of 1.00 mol/dm^3 sodium hydroxide (NaOH) with 100 cm^3 of 1.00 mol/dm^3 sulfuric acid.

END OF CHEMISTRY ONLY